



# The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

**1796 – 2020**

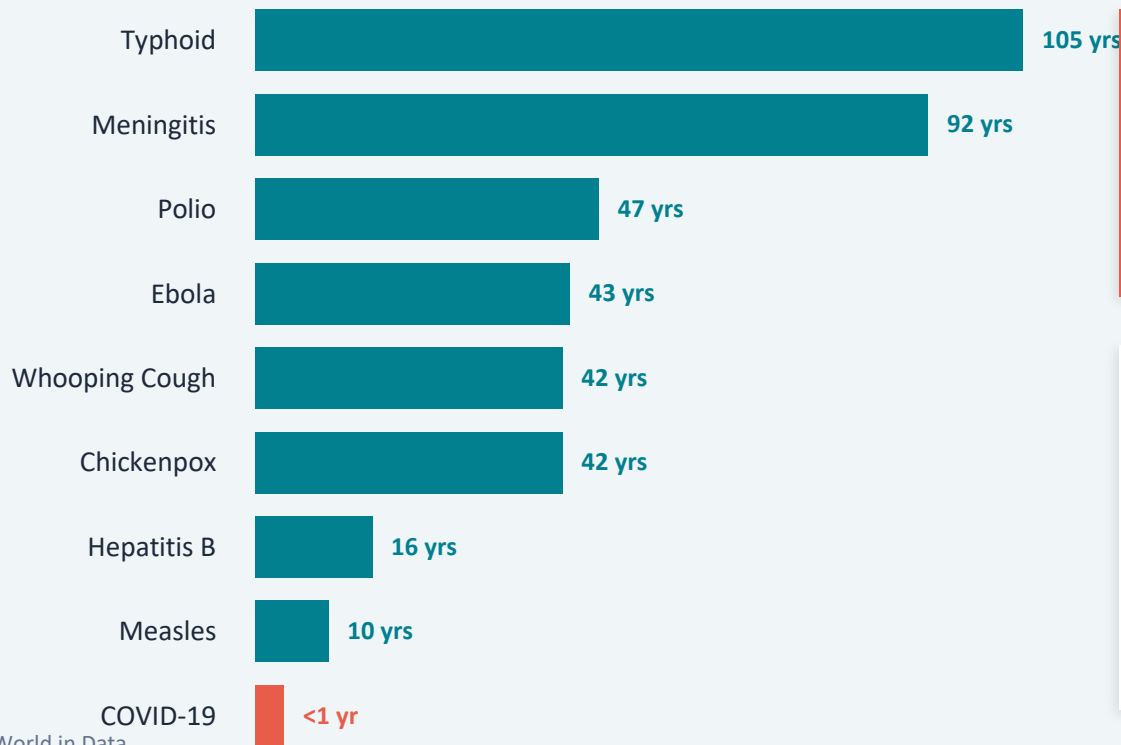
A data-driven overview of how vaccines have transformed public health by eliminating or drastically reducing deadly diseases over two centuries.

Source: Our World in Data — [ourworldindata.org/vaccination](https://ourworldindata.org/vaccination)



# From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure



Source: Our World in Data

COVID-19

<1 year

Pathogen to licensure

## Key Driver

mRNA technology, genome sequencing, and advanced microscopy have compressed development from over a century to months.



# 100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

| Disease      | Pre-Vaccine Cases/million/yr | Post-Vaccine Cases/million/yr | Case Reduction | Death Reduction |
|--------------|------------------------------|-------------------------------|----------------|-----------------|
| Diphtheria   | 158                          | 0                             | 100%           | 100%            |
| Smallpox     | 250                          | 0                             | 100%           | 100%            |
| Acute Polio  | 141                          | 0                             | 100%           | 100%            |
| Measles      | 3,044                        | 0.2                           | 99.99%         | 100%            |
| Rubella      | 242                          | 0.04                          | 99.98%         | 100%            |
| Pertussis    | 1,534                        | 52                            | 96.6%          | 99.7%           |
| Hepatitis A  | 465                          | 51                            | 89%            | 88.7%           |
| Hepatitis B  | 273                          | 44                            | 83.9%          | 83.6%           |
| Pneumococcal | 233                          | 139                           | 40.5%          | 31.3%           |

100% elimination (cases and/or deaths)

Incomplete reduction — gaps remain

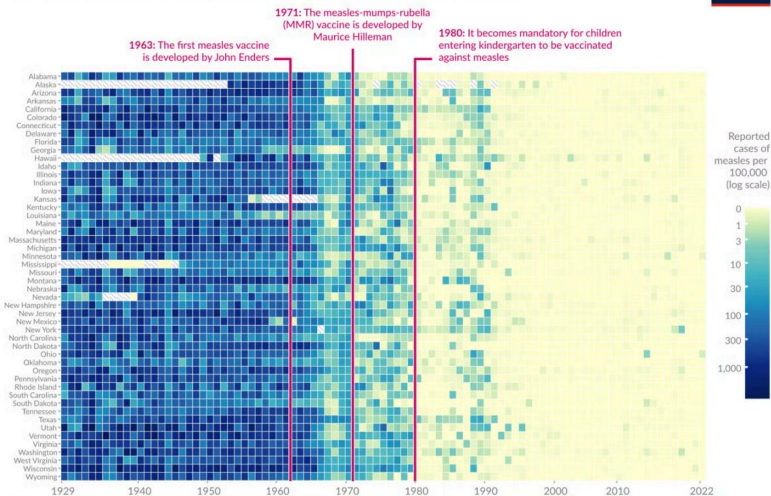
Source: Roush & Murphy (2007), via Our World in Data



# Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

## Vaccines reduced measles cases across US states

Our World  
in Data



Data source: Project Tycho (2018); Centers for Disease Control and Prevention (1959–2022)

OurWorldinData.org — Research and data to make progress against the world's largest problems.

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Download image or data

1963

First measles vaccine developed by John Enders

1971

MMR (measles-mumps-rubella) vaccine developed by Maurice Hilleman

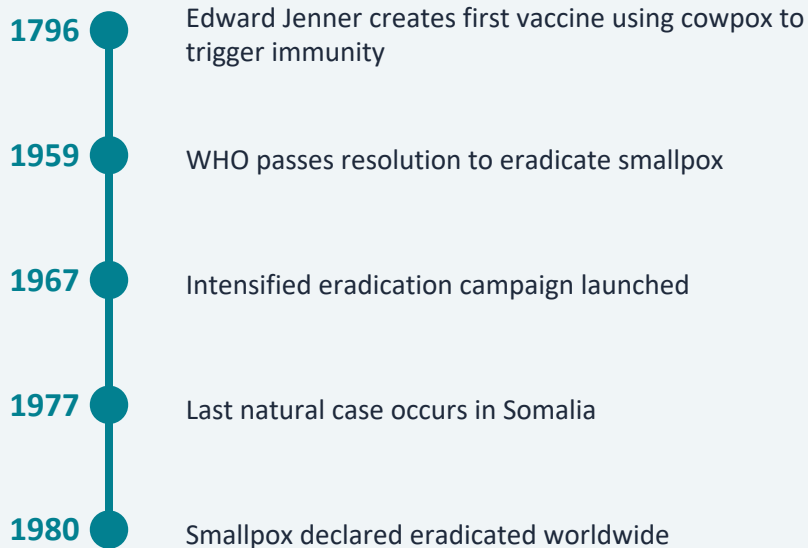
1980

Vaccination became mandatory for children entering kindergarten

Before 1963: 100–1,000+ cases per 100,000. After 1980: near zero — only sporadic outbreaks.



# Smallpox: The Only Human Disease Eradicated Through Vaccination



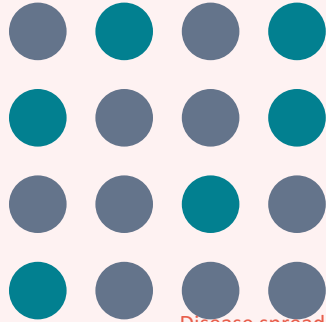
## Why smallpox was uniquely suited for eradication

- ✓ The virus infected only humans — no animal reservoir
- ✓ Symptoms were visually distinguishable, enabling rapid identification
- ✓ The vaccine was cheap, stable, and highly effective



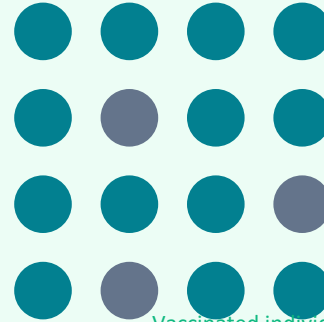
# Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

## Low Vaccination



Disease spreads easily  
through unvaccinated population

## High Vaccination (Herd Immunity)



Vaccinated individuals shield  
vulnerable community members



Vaccinated



Unvaccinated

Vulnerable groups protected: newborns • immunocompromised patients • cancer patients on chemotherapy

When enough people are vaccinated, pathogens cannot spread effectively, shielding those who cannot be vaccinated.



# Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

## Pneumococcal Disease

40.5%

case reduction



Death reduction: 31.3%

*vs. 100% for diphtheria, polio, smallpox*

## Hepatitis A

89%

case reduction



Death reduction: 88.7%

*vs. 100% for diphtheria, polio, smallpox*

## Hepatitis B

83.9%

case reduction



Death reduction: 83.6%

*vs. 100% for diphtheria, polio, smallpox*

## Barriers to full coverage

- Access gaps — not everyone has access to vaccines globally
- Supply shortages — put children at risk in underserved regions
- Trust challenges — poor scientific understanding can reduce uptake

# Key Takeaways

- 1 Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.
- 2 Development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.
- 3 Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.
- 4 Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign.
- 5 Gaps persist for pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%).

**Call to Action:** Improve access, coverage, and trust in vaccination to close remaining gaps and protect global health systems.